Linux DHCP Server Implementation

NETWORK & SYSTEM ADMINISTRATION | openSUSE + VirtualBox | August 2026

Portfolio case study | Chia Li Ying | Academic team project

Objective & context

Build an isolated 192.168.2.0/24 Linux lab LAN in which DHCP automatically provisions client network settings. The wider team environment included Apache web hosting, BIND DNS and a wireless access point; the personal contribution documented here is DHCP deployment and validation.

Architecture and addressing

Component	Address / Range	Function
openSUSE DHCP server	192.168.2.1/24	DHCP address allocation
DNS server (team)	192.168.2.20	Provided as DNS option
Web server (team)	192.168.2.10	Static host, outside dynamic pool
Client pool	192.168.2.100 - .200	Dynamic DHCP leases
Reserved client	192.168.2.50	MAC-bound fixed assignment

My contributions

- **Server configuration:** Set the static server address using nmcli; configured DHCP interface binding, /etc/dhcpd.conf, subnet/pool parameters, router and DNS options, and lease times.
- **IP reservation:** Created a fixed DHCP address mapping for a test VM using its MAC address, outside the dynamic range.
- **Testing & troubleshooting:** Used dhcpd -t -cf /etc/dhcpd.conf and systemctl to verify service configuration and operation. Checked client leases and resolver settings; corrected a subnet declaration error during setup.

Verification evidence recorded in the group report

- **Dynamic assignment:** Client VM obtained 192.168.2.100/24 from the configured range; /etc/resolv.conf pointed at the group DNS server.
- **Static reservation:** MAC-bound test client obtained reserved 192.168.2.50 rather than an address from the dynamic pool.
- **Availability:** DHCP service was reported active; configured to start on boot via systemctl enable dhcpd.

Tools & technical concepts

openSUSE Linux • VirtualBox • NetworkManager / nmcli • ISC DHCP (dhcpd) • IPv4 subnetting • DNS option assignment • MAC/IP reservation • systemctl • client verification